

Problem L. Longest Common Prefix

Input file: **standard input**
Output file: **standard output**
Time limit: **1 second**
Memory limit: **1024 megabytes**

You are given n lowercase English strings $s_1, s_2, \dots, s_n$.

For each pair of integers (i, j) satisfying $1 \leq j \leq i \leq n$, define $f_{i,j}$ as the maximum possible length of the longest common prefix of exactly j strings selected from the first i strings. Formally,

$$f_{i,j} = \max_{T \subseteq \{1, 2, \dots, i\}, |T|=j} |\text{LCP}(s_k \mid k \in T)|.$$

The longest common prefix of a single string is the string itself.

For each $i = 1, 2, \dots, n$, calculate

$$\sum_{j=1}^i (f_{i,j} \oplus j),$$

where $\oplus$ denotes the bitwise XOR operation.

Input

The first line contains an integer n ($1 \leq n \leq 5 \cdot 10^5$), the number of strings.

Each of the next n lines contains a non-empty string s_i consisting only of lowercase English letters.

It is guaranteed that

$$\sum_{i=1}^n |s_i| \leq 5 \cdot 10^5.$$

Output

Print n lines.

The i -th line should contain one integer:

$$\sum_{j=1}^i (f_{i,j} \oplus j).$$

Examples

standard input	standard output
4 a ab abc abcd	0 6 4 12
8 ab b ba ab aa ba a bb	3 5 9 10 14 20 28 36

Note

The longest common prefix of several strings is the longest string that is a prefix of every selected string.

The symbol $\oplus$ denotes the bitwise XOR operation.